



Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 08

Team No.: 09

Project Title: Linux System Information and Resource Monitoring Tool

Team Members

Roll Number	Student Name	Signature
2520030027	Venkata Sai Krishna	
2520030029	Pujith Pavan Kumar	
2520030320	Ajai Manikanta	
2520030437	Bandla Vinay	

1. Abstract

Abstract

The Linux System Information and Resource Monitoring Tool is a simple Linux-based project developed to monitor and display important information about a computer system. In a Linux environment, several processes run at the same time and continuously use resources such as the CPU, memory, and storage. Understanding how these resources are being used is useful for identifying performance problems and managing the system efficiently. The project is implemented using C programming and Linux system programming concepts. It collects system information using Linux interfaces such as the /proc filesystem and system calls. The project covers important Operating Systems concepts including process management, memory management, file systems, OS services, inter-process communication, and concurrency.

The proposed tool provides a simple menu-driven interface where users can view CPU utilization, memory usage, disk space, running processes, system uptime, and other basic system information. The information is presented in a clear and easy-to-understand manner.

The main aim of this project is to connect the theoretical concepts learned in the Operating Systems syllabus with practical Linux programming. The expected outcome is a lightweight and user-friendly monitoring tool that provides useful system information while helping students understand how Linux manages processes and system resources in real time.

2. Problem Statement

Linux systems run multiple processes at the same time and use resources such as CPU, memory, disk space, and storage. For beginners, it can be difficult to understand how these resources are being used and identify processes that may affect system performance. Using different commands separately can also make system monitoring inconvenient.

This project aims to develop a simple Linux System Information and Resource Monitoring Tool that displays important system information in one place. It is relevant to Operating Systems and Systems Programming as it applies concepts such as process management, memory management, file systems, system calls, and resource management using C programming and Linux system interfaces.

3. Objectives

1. To monitor CPU, memory, disk usage, and running processes in a Linux system.
2. To collect and display system information using Linux system interfaces such as /proc.
3. To apply Operating System concepts like process management, memory management, file systems, and system calls.
4. To develop a simple and user-friendly C-based tool for understanding and monitoring system resources.

4. Proposed Methodology

The project will be developed using C programming in a Linux environment. The program will collect system information such as CPU usage, memory usage, disk space, running processes, and system uptime using Linux interfaces like the /proc filesystem and system calls.

The tool will have a simple menu-based interface so that users can easily select what information they want to see. Based on the user's choice, the program will collect the required data, process it, and display the results clearly. The basic flow will be User Selection → Collect System Data → Process Data → Display Results. Finally, the tool will be tested with different system conditions to make sure the information shown is correct and useful.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	To monitor and display currently running processes.
/proc Filesystem	To collect CPU, memory, uptime, and process information from the Linux system.
System Calls (open(), read(), close())	To access and read system-level information from Linux files.
Memory Management	To monitor total, used, and available memory.
File System / Disk Management	To check disk space and storage usage.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520030027	Venkata Sai Krishna	Process monitoring and system information module
2520030029	Pujith Pavan Kumar	Project planning, system information module, and overall integration
2520030320	Ajai Manikanta	CPU and memory monitoring module
2520030437	Bandla Vinay	Disk monitoring, testing, and documentation

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating environment for developing and running the project.
C Programming	Used to develop the system monitoring application.
GCC Compiler	Used to compile and run the C program.
VS Code / Text Editor	Used for writing and editing the source code.
Linux Terminal	Used to compile, execute, and test the project.

8. Expected Outcome

The project is expected to provide a simple and user-friendly Linux monitoring tool that displays important system information in one place. The user will be able to check CPU usage, memory usage, disk space, running processes, system uptime, and basic system details through a menu-driven interface.

The tool will collect and process system information using C programming, Linux system calls, and the /proc filesystem. The expected result is a lightweight and reliable application that helps users understand their system performance and demonstrates practical Operating Systems concepts such as process management, memory management, file systems, and resource monitoring.

9. GitHub Repository

GitHub Repository Link: https://github.com/V-max-OSSP/V-Max_OSSP_Project

Repository Created: Yes

Faculty Name and Signature

Faculty Name: _____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____